

Homeless Services Outcome Analysis

A data analytics portfolio project evaluating the effectiveness of homeless services programs and identifying the primary drivers of successful permanent housing placement.

Overview

Nonprofit organizations providing homeless services operate with tight budgets and face the constant challenge of placing vulnerable individuals into permanent housing. Frontline agencies must make critical resource allocation decisions, yet they often lack quantitative evidence on which interventions most effectively drive housing stability.

This project bridges the gap between administrative data and strategic operations. By analyzing client pathways, program durations, and case management models, this analysis provides nonprofit leadership with evidence-based recommendations. It identifies the true operational drivers of successful housing transitions, allowing leaders to optimize staffing levels, target capital investments, and back grant requests with rigorous, databacked findings.

Dataset

The analysis uses a refined administrative dataset consisting of client intake, engagement, and exit records.

- * **Source:** Simulated administrative records reflecting real-world homeless services operations.
- * **Size:** 150 client records, 20 columns.
- * **Key Variables:**
 - * ``exit_status`` (Primary Outcome): Categorized as `*Permanent*` (successful placement) or `*Other*` (temporary housing, shelter, or return to homelessness).
 - * **Demographics:** ``age``, ``gender``, ``veteran_status``, ``disability_status``.
 - * **Client History:** ``prior_episodes`` (number of prior homeless episodes in the last 3 years).
 - * **Programmatic & Operations:** ``program_type`` (Emergency Shelter, Rapid Rehousing, Transitional Housing, Permanent Supportive Housing), ``length_of_stay`` (days enrolled), ``services_received`` (count of unique services), ``case_management_intensity`` (Low, Medium, High).

Project Structure

```
data/
├── homeless_services_raw.csv      # Raw messy administrative data
homeless_services_clean.csv      # Cleaned, standardized dataset
scripts/
├── audit.R                       # Data validation, screening, and profiling
├── clean.R                       # Data ingestion, type correction, and cleaning
├── eda.R                         # Exploratory visualizations and summary stats
modeling.R                       # Primary logistic and secondary linear regression
report/
├── homeless_services_analysis.qmd # Quarto report combining code and narrative
├── homeless_services_analysis.html # Rendered, interactive HTML dashboard
├── project_framing_and_analysis_plan.md # Framing and analysis methodology plan
├── data_intake_and_screening.md    # Detailed initial data screening observations
└── data_cleaning_report.md        # Cleaning operations log and before/after details
```

```

├── modeling_analysis.md          # Detailed statistical modeling and interpretation
└── executive_summary_printable.pdf # Printable PDF summary for executive leadership
outputs/
├── logit_results.csv            # Logistic regression model coefficients
├── linear_results.csv           # Linear regression model coefficients
├── model_performance_table.csv   # Confusion matrix for primary model
├── model_linear_diagnostics.png  # Diagnostic plots for the linear model
└── figures/                     # Explanatory charts from EDA
    ├── 02_numeric_vars.png
    ├── 03_categorical_vars.png
    ├── 04_outcome_vs_program.png
    ├── 05_outcome_vs_stay.png
    └── 06_correlation.png
README.md                        # Main project portfolio description
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```

Methodology

1. Data Screening

Conducted structural checks to ensure primary key (`client\_id`) integrity, validate date ranges, check for negative numbers or extreme values, and identify missing data patterns (e.g., missingness in `housing\_placement\_days` and `monthly\_income\_at\_exit`).

2. Data Cleaning

- * Standardized inconsistent text formatting in `program\_type` (e.g., double spacing and mixed capitalization like "Transitional housing").
- * Addressed missing variables using logical rules: imputed missing `housing\_placement\_days` for clients exiting to *Permanent* housing using their total program `length\_of\_stay`. * Retained NA values for sensitive columns like income to prevent model bias.

3. Exploratory Data Analysis (EDA) * Analyzed distribution of housing outcomes to identify baseline success rates.

- * Evaluated bivariate relationships, highlighting the success rate of various program types (e.g., PSH vs. Emergency Shelter) and length of stay variations.
- * Generated correlation heatmaps of numeric variables to identify operational relationships and check for potential multicollinearity.

4. Statistical Modeling

- * **Primary Model (Logistic Regression):** Modeled the log-odds of a successful *Permanent* housing exit using client age, stay length, program type, and income at entry, calculating odds ratios to isolate programmatic effects while controlling for baseline demographics.
- * **Secondary Model (Multiple Linear Regression):** Analyzed the predictors of program stay duration (`length\_of\_stay`), revealing that program assignment acts as the primary constraint on time in program.
- * **Model Diagnostics:** Evaluated assumptions using residual plots to ensure model validity.

Key Insights

- * **Case Management is the Primary Actionable Lever:** Operational data shows that highintensity case management is heavily associated with successful permanent housing exits. Lowering intensity to "Medium" or "Low" significantly reduces placement success.
- * **PSH is the Gold Standard for Stabilization:** Permanent Supportive Housing (PSH)

achieves a near-perfect housing placement rate of **96.4%**, though it requires longer program durations (averaging 239 days).

- * **Transitional Housing (TH) and Rapid Rehousing (RRH) offer High Efficiency:** Both models achieve high placement rates (**82.7%** and **77.8%** respectively) with a much shorter duration (~122 days for TH, ~100 days for RRH), proving highly cost-effective.
- * **Time Equals Stability:** Longer lengths of stay are linked to permanent placements, indicating that rushing individuals out of programs increases the risk of temporary exits or returns to homelessness.
- * **Attrition Risk in Chronically Unhoused Clients:** Clients entering with high numbers of prior homeless episodes tend to have shorter stay durations, highlighting a risk of early dropout.

Tools Used

- Antigravity/Gemini
- Chatgpt
- Rstudio
- R
- Quarto
- Tidyverse - several libraries

How to Run

Central to the analysis is a 9 prompt workflow to be used in Antigravity/Gemini, run one prompt at a time.

Prompts are:

01_project_overview.md

02_data_screening_master.md

03_data_cleaning_master.md

04_eda_master.md

05_modeling_master.md

06_quarto_report_master.md

07_readme_master.md

08_code_review_master.md

09_final_qa_packaging.md

Prerequisites

- Ensure you have R, RStudio, and Quarto installed on your machine.
- Install Required R Packages Open an R console and run the following command to install the required packages:

```
```r
```

- `install.packages(c("tidyverse", "broom", "skimr", "here", "janitor"))`

...

### Output from the 9 prompt workflow, each prompt run individually.

- Each of the 9 prompts run in Antigravity/Gemini will have some output and analysis, to be saved and possibly printed. See the folder structure for specific outputs.
- Prompts 02-05 also output rscript generated by Antigravity/Gemini.
  - Prompt 02 is for data screening and generates rscript file scripts/audit.R.
  - Prompt 03 is for data cleaning and generates rscript file scripts/clean.R.
  - Prompt 04 is for EDA and generates rscript file scripts/eda.R.
  - Prompt 05 is for Modeling and generates rscript file scripts/modeling.R.
- Each of the 4 rscripts are run and debugged in Rstudio, one at a time, to check the integrity of the coding. Output from running the 4 rscript files in Rstudio is examined and saved, if appropriate. This output includes that from the console in Rstudio.
- After debugging each of the 4 rscripts in Rstudio, each individual script is further analyzed in chatgpt for errors and possible code improvements. A prompt named "segmented R screening prompt.md" is used to evaluate each rscript as it comes out of Rstudio debugging. Rscript is pasted into the segmented prompt. If needed, changes are made to the code and each is rerun in Rstudio, if modified.
- At this point, each of the remaining prompts, 06 thru 09 are run, one at a time.
- After running prompt 06 render the Quarto Report,.qmd, to render the final HTML report. Report is  
"report/homeless\_services\_analasis.html"

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